



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSA05-DATABASE MANAGEMENT SYSTEMS

LAB MANUAL

DDL Commands – CREATE, ALTER, DROP

AIM:

To Create, Alter, Drop, Rename, Truncate using Data Definition Language (DDL) statements.

Description:

Data Definition Language (DDL) statements are used to define the database structure or schema.

DDL Commands: Create, Alter, Drop, Rename, Truncate

- CREATE - to create objects in the database
- ALTER - alters the structure of the database
- DROP - delete objects from the database
- TRUNCATE - remove all records from a table, including all spaces allocated for the records are removed
- RENAME - rename an object

SYNTAX:

CREATE TABLE

CREATE TABLE table_name

```
(  
column_name1 data_type,  
column_name2 data_type,  
column_name3 data_type,  
....  
);
```

ALTER A TABLE

To add a column in a table

```
ALTER TABLE table_name  
ADD column_namedatatype;
```

To delete a column in a table

ALTER TABLE table_name

DROP COLUMN column_name;

DROP TABLE

DROP TABLE table_name;

TRUNCATE TABLE

TRUNCATE TABLE table_name;

Questions:

1) Create a table name STUDENT with following structure.

#	Column Name	Description	Data Type
1	RegNo	Registration Number	NUMBER(3)
2	Name	Student Name	VARCHAR(15)
3	Gender	Gender of the student	CHAR(1)
4	DOB	Date of Birth	DATE
5	MobileNo	Mobile Number	NUMBER(10)
6	City	Location of stay	VARCHAR(15)

2) Create a table name FACULTY with following structure.

	Column #Name	Description	Data Type
1	FacNo	Faculty Identifier	VARCHAR(4)
2	FacName	Faculty Name	VARCHAR(15)
3	Gender	Gender of faculty	CHAR(1)
4	DOB	Date of Birth	DATE
5	DOJ	Date of Join	DATE
6	MobileNo	Mobile Number	NUMBER(10)

3) Create a table name DEPARTMENT with following structure.

	Column #Name	Description	Data Type
1	DeptNo	Department Identifier	VARCHAR(4)
2	DeptName	Department Name	VARCHAR(15)
3	DeptHead	Department Head	VARCHAR(4)

4) Create a table name COURSE with following structure.

	Column #Name	Description	Data Type
1	CourseNo	Course Identifier	VARCHAR(3)
2	CourseDesc	Course Description	VARCHAR(14)
3	CourseType	Course Type	CHAR(1)
4	SemNo	Semester Number	CHAR(1)
5	HallNo	Hall Number	VARCHAR(4)
6	FacNo	Faculty Identifier	VARCHAR(4)

5) Modify the table FACULTY by adding a column name Dept of datatype VARCHAR(10)

OUTPUTS:

1)

```
mysql> CREATE DATABASE college_dd;
Query OK, 1 row affected (0.288 sec)

mysql> USE college_dd;
Database changed
mysql> CREATE TABLE STUDENT2
-> (
->     RegNo INT,
->     S1Name VARCHAR(30)
-> );
Query OK, 0 rows affected (0.494 sec)

mysql> ALTER TABLE STUDENT2
-> ADD PRIMARY KEY (RegNo);
Query OK, 0 rows affected (0.391 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc STUDENT2;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int           | NO   | PRI | NULL    |       |
| S1Name | varchar(30)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.032 sec)
```

2)

```
mysql> CREATE TABLE FACULTY
-> (
->     FacNo INT,
->     FacName VARCHAR(30),
->     Gender CHAR(1)
-> );
Query OK, 0 rows affected (0.476 sec)
```

```
mysql>
mysql> ALTER TABLE FACULTY
-> ADD PRIMARY KEY (FacNo);
Query OK, 0 rows affected (0.568 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> ALTER TABLE FACULTY
-> ADD CONSTRAINT chk_gender
-> CHECK (Gender IN ('M','F'));
Query OK, 0 rows affected (0.654 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc FACULTY;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| FacNo | int           | NO   | PRI | NULL    |       |
| FacName | varchar(30)  | YES  |     | NULL    |       |
| Gender | char(1)      | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.025 sec)
```

3)

```
mysql> CREATE TABLE DEPARTMENT
-> (
->     DeptNo INT,
->     DeptName VARCHAR(30)
-> );
Query OK, 0 rows affected (0.467 sec)

mysql>
mysql> ALTER TABLE DEPARTMENT
-> ADD PRIMARY KEY (DeptNo);
Query OK, 0 rows affected (0.501 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc DEPARTMENT;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| DeptNo     | int           | NO   | PRI | NULL    |       |
| DeptName   | varchar(30)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.030 sec)
```

4)

```
mysql> CREATE TABLE COURSE
-> (
->     CourseNo INT,
->     CourseName VARCHAR(30),
->     SemNo INT
-> );
Query OK, 0 rows affected (0.295 sec)
```

```
mysql>
mysql> ALTER TABLE COURSE
-> ADD PRIMARY KEY (CourseNo);
Query OK, 0 rows affected (0.561 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> ALTER TABLE COURSE
-> ADD CONSTRAINT chk_sem
-> CHECK (SemNo BETWEEN 1 AND 6);
Query OK, 0 rows affected (1.003 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc COURSE;
```

Field	Type	Null	Key	Default	Extra
CourseNo	int	NO	PRI	NULL	
CourseName	varchar(30)	YES		NULL	
SemNo	int	YES		NULL	

3 rows in set (0.022 sec)

5)

```
mysql> ALTER TABLE STUDENT2
-> ADD CONSTRAINT uk_s1name
-> UNIQUE (S1Name);
Query OK, 0 rows affected (0.493 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc STUDENT2;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	
S1Name	varchar(30)	YES	UNI	NULL	

2 rows in set (0.021 sec)

RESULT:

Tables are created, altered and modified using DDL commands.